Mathematical Modeling and Optimization
Final Test — Structure, Guidelines, and Sample Questions

1. General Information and Exam Rules

The Final Test evaluates your understanding of the theoretical foundations of classical optimization, linear regression models, and optimal experimental design, as well as your ability to solve practical problems based on the homework assignments.

- **Time Limit:** 120 minutes (2 hours).
- **Maximum Score: 50 points** in total.
- **Format:** Closed-book, in-person written examination. No electronic devices, internet access, lecture notes, or reference materials are permitted during the test.
- **Test Materials:** Copying or photographing the test questions is strictly prohibited. You will be able to review your graded work in the presence of the instructor.

Overview of the Test Structure

Section	Question Type	Number of Tasks	Points
Block 1	Multiple-Choice Questions (MCQs)	10 questions × 1 point	**10 points**
Block 2	Short-Answer Theoretical Questions	8 questions × 3 points	**24 points**
Block 3	Problem-Solving Exercises	4 problems (2 × 3 pts + 2 × 5 pts)	**16 points**
Total		**22 tasks**	**50 points**

2. Detailed Guide to Each Block & Sample Questions

Block 1 — Multiple-Choice Questions (10 points in total)

What to expect: Block 1 consists of **10 multiple-choice questions**, worth **1 point each**. The questions cover basic definitions, properties of algorithms, optimality conditions, duality, least squares estimation, and optimality criteria in experimental design.

What you need to do: For each question, four options (a, b, c, d) are provided, and **only one option is correct**. You only need to select (circle or write down) the single correct letter. No written justification is required in Block 1.

Sample Question for Block 1 (1 point):

The constraint qualification called LICQ (Linear Independence Constraint Qualification) at a feasible point x^* requires that:

a) The gradients of active constraints at x^* are linearly independent
b) All constraints in the problem are linear functions
c) The objective function and all constraints are linear
d) The gradients of all constraints (both active and inactive) at x^* are linearly independent

Model Answer:

- **Correct Answer:** a)

Block 2 — Short-Answer Theoretical Questions (24 points in total)

What to expect: Block 2 contains 8 theoretical questions, worth 3 points each. Each question asks you to state precise mathematical definitions, formulas, lemmas, or theorems and answer one or more short conceptual questions. The number of sub-questions and the allocation of points may vary from one task to another.

What you need to do: When stating theorems or definitions, pay close attention to mathematical precision: specify all required assumptions, including differentiability, positive definiteness, or constraint qualifications where relevant, and define all symbols you use. Proofs are not required unless explicitly requested. For conceptual questions, provide a concise mathematical justification whenever appropriate.

Sample Question for Block 2 (3 points):

Moré–Sorensen theorem for the trust-region subproblem (3 points)
Consider the trust-region subproblem

$$\min_{p\in\mathbb{R}^n} m(p) = f + g^\top p + \frac{1}{2}p^\top Bp \quad \text{s.t.} \quad \|p\| \le \Delta,$$

where $g \in \mathbb{R}^n$, $B \in \mathbb{R}^{n\times n}$ is symmetric, and $\Delta > 0$ is the trust-region radius.

(a) **(2 points)** State precisely the **Moré–Sorensen theorem** (Theorem 12) that characterizes the global solution p^* of this subproblem.

(b) **(1 point)** *Question:* Suppose B is symmetric positive definite and the unconstrained minimizer $p^B = -B^{-1}g$ satisfies $\|p^B\| < \Delta$. What is the exact value of the scalar multiplier λ in the Moré–Sorensen optimality conditions, and why?

Model Answer & Grading Breakdown:

- **Part (a) [2 points]: Theorem (Moré–Sorensen):** The vector p^* is a global solution of the trust-region subproblem

$$\min_{p\in\mathbb{R}^n} m(p) = f + g^\top p + \frac{1}{2}p^\top Bp \quad \text{s.t.} \quad \|p\| \le \Delta$$

if and only if p^* is feasible ($\|p^*\| \le \Delta$) and there exists a scalar $\lambda \ge 0$ such that the following three conditions hold:

$$(B + \lambda I)p^* = -g, \tag{1}$$
$$\lambda(\Delta - \|p^*\|) = 0, \tag{2}$$
$$(B + \lambda I) \text{ is positive semidefinite.} \tag{3}$$

(Grading note: Full 2 points require stating feasibility $\|p^*\| \le \Delta$, nonnegativity $\lambda \ge 0$, and all three conditions (1)-(3). Omitting positive semidefiniteness of $B+\lambda I$ or complementarity $\lambda(\Delta-\|p^*\|) = 0$ results in partial credit.)

- **Part (b) [1 point]: Answer:** $\lambda = 0$.
Justification: Since $\|p^B\| < \Delta$, the global solution is $p^* = p^B = -B^{-1}g$, which lies strictly inside the trust region ($\Delta - \|p^*\| > 0$). By the complementarity condition $\lambda(\Delta - \|p^*\|) = 0$, we must have $\lambda = 0$.

Block 3 — Problem-Solving Exercises (16 points in total)

What to expect: Block 3 consists of **4 practical problems**:

- **Problems 1 and 2:** Single-part problems worth **3 points each** (6 points total).
- **Problems 3 and 4:** Two-part problems (with sub-questions *(a)* and *(b)*, worth 2.5 points per sub-question) worth **5 points each** (10 points total).

IMPORTANT NOTE ON HOMEWORK PROBLEMS:

All problems in Block 3 are directly based on the homework exercises you solved during the semester. **However, the problems on the exam will NOT be identical word-for-word copies of the homework!** They will have **different numerical coefficients, shifted parameters, or slightly modified conditions**.

Memorizing final answers from the homework will not help you and trying to "force" a solution to match a memorized homework answer will result in zero points. Make sure you understand the underlying solution methods and derivations!

What you need to do: Write down clear, step-by-step mathematical derivations and justifications for every step of your solution.

Sample Problem for Block 3 (5 points):

(Compare this with Homework 5, Exercise 6 — notice how the objective function and constraint parameters have been modified while keeping the exact same mathematical structure.)

Consider the constrained optimization problem:

$$\min_{x\in\mathbb{R}^2} f(x) = -3x_1 + x_2 \quad \text{subject to} \quad \begin{cases} c_1(x) = (1-x_1)^3 - x_2 + 1 \geq 0, \\ c_2(x) = x_2 + 0.5x_1^2 - 2 \geq 0. \end{cases}$$

Consider the feasible point $x^* = (0,2)^\top$, where both constraints are active.

(a) **(2.5 points)** Check whether the LICQ holds at $x^* = (0,2)^\top$ and verify whether the first-order KKT conditions are satisfied at x^* (find the exact Lagrange multipliers λ_1^*, λ_2^*).

(b) **(2.5 points)** Write down the linearized feasible direction set $\mathcal{F}(x^*)$ and the critical cone $\mathcal{C}(x^*, \lambda^*)$, and check whether the second-order necessary and sufficient conditions hold at x^*.

Model Solution & Grading Breakdown:

Part (a) [2.5 points]:

1. **Check activity and compute gradients (1 point):** At $x^* = (0,2)^\top$, we have $c_1(0,2) = (1-0)^3 - 2 + 1 = 0$ and $c_2(0,2) = 2 + 0 - 2 = 0$, so both constraints are active: $\mathcal{A}(x^*) = \{1,2\}$. Computing the gradients of f, c_1, and c_2:

$$\nabla f(x) = \begin{bmatrix} -3 \\ 1 \end{bmatrix}, \qquad \nabla c_1(x) = \begin{bmatrix} -3(1-x_1)^2 \\ -1 \end{bmatrix}, \qquad \nabla c_2(x) = \begin{bmatrix} x_1 \\ 1 \end{bmatrix}.$$

   Evaluating at $x^* = (0,2)^\top$:

$$\nabla f(x^*) = \begin{bmatrix} -3 \\ 1 \end{bmatrix}, \qquad \nabla c_1(x^*) = \begin{bmatrix} -3 \\ -1 \end{bmatrix}, \qquad \nabla c_2(x^*) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

2. **Verify LICQ (0.5 points):** The matrix of active constraint gradients is $\begin{bmatrix} \nabla c_1(x^*) & \nabla c_2(x^*) \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ -1 & 1 \end{bmatrix}$. Its determinant is $-3 \neq 0$, so $\nabla c_1(x^*)$ and $\nabla c_2(x^*)$ are linearly independent. Thus, **LICQ holds at x^***.

3. **Verify KKT conditions (1 point):** Stationarity $\nabla_x \mathcal{L}(x^*, \lambda^*) = \nabla f(x^*) - \lambda_1^* \nabla c_1(x^*) - \lambda_2^* \nabla c_2(x^*) = 0$ yields the system:

$$\begin{bmatrix} -3 \\ 1 \end{bmatrix} - \lambda_1^* \begin{bmatrix} -3 \\ -1 \end{bmatrix} - \lambda_2^* \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{cases} -3 + 3\lambda_1^* = 0, \\ 1 + \lambda_1^* - \lambda_2^* = 0. \end{cases}$$

   Solving gives $\lambda_1^* = 1$ and $\lambda_2^* = 2$. Since $\lambda_1^* \geq 0$ and $\lambda_2^* \geq 0$, all KKT conditions (primal feasibility, dual feasibility, complementarity, and stationarity) are **satisfied**.

Part (b) [2.5 points]:

1. **Find $\mathcal{F}(x^*)$ and $\mathcal{C}(x^*, \lambda^*)$ (1.5 points):** By definition, for $d = (d_1, d_2)^\top \in \mathbb{R}^2$:

$$\mathcal{F}(x^*) = \left\{ d \in \mathbb{R}^2 \,\middle|\, \nabla c_1(x^*)^\top d \geq 0,\ \nabla c_2(x^*)^\top d \geq 0 \right\} = \left\{ (d_1, d_2)^\top \in \mathbb{R}^2 \,\middle|\, -3d_1 - d_2 \geq 0,\ d_2 \geq 0 \right\}.$$

Since both Lagrange multipliers are strictly positive ($\lambda_1^* = 1 > 0$ and $\lambda_2^* = 2 > 0$), the critical cone $\mathcal{C}(x^*, \lambda^*)$ requires equality for both active constraints:

$$\mathcal{C}(x^*, \lambda^*) = \left\{ d \in \mathcal{F}(x^*) \,\middle|\, \nabla c_1(x^*)^\top d = 0,\ \nabla c_2(x^*)^\top d = 0 \right\} = \left\{ (d_1, d_2)^\top \,\middle|\, -3d_1 - d_2 = 0,\ d_2 = 0 \right\} = \{(\mathbf{0}, \mathbf{0})^\top\}.$$

2. **Check second-order conditions (1 point):**
   - The second-order **necessary** condition requires $w^\top \nabla^2_{xx} \mathcal{L}(x^*, \lambda^*) w \geq 0$ for all $w \in \mathcal{C}(x^*, \lambda^*)$. Since $\mathcal{C}(x^*, \lambda^*) = \{0\}$, for $w = 0$ we have $0 \geq 0$, so it is **trivially satisfied**.
   - The second-order **sufficient** condition requires $w^\top \nabla^2_{xx} \mathcal{L}(x^*, \lambda^*) w > 0$ for all $w \in \mathcal{C}(x^*, \lambda^*) \backslash \{0\}$. Since the set $\mathcal{C}(x^*, \lambda^*) \setminus \{0\} = \emptyset$ is empty, the condition holds **vacuously (is satisfied)**, confirming that $x^* = (0, 2)^\top$ is a strict local minimizer.